

1.1 Introduction to Set

Review

- Sets = collection of objects
- Objects are entities such as integers, alphabet symbols etc.

For example, the collection of the three letters a , b , and c is a set, which we may name A ; we write $A = \{a, b, c\}$.

- Equality

Two set are equal (that is, the same) if any only if they have the same elements.

$$A \equiv \{a, b, c\} \equiv \{b, a, c, a\}$$

- Membership

denoted by $a \in A$, we read a is in A .

Review

- Super-set, complement

If the set E is English alphabet, then

$$A = \{\alpha \mid \text{first 3 letters of the alphabet } E\}.$$

In this, case A is a subset of E and E is the super-set of A ,

$$\text{i.e. } A \subseteq E \text{ and } E \supseteq A.$$

$$\text{And, } A' = \{\alpha \subset E \mid \alpha \notin A\}$$

is the complement of A , i.e., elements other than a , b or c .

- Singleton $A = \{1\}$

- Infinite $N = \{0, 1, 2, \dots\}$

- Empty $A = \{\}$ or $\varnothing$

Set Operations

- Union

$$A \cup B = \{x: x \in A \text{ or } x \in B\}.$$

For example,

$$\{1, 3, 9\} \cup \{3, 5, 7\} = \{1, 3, 5, 7, 9\}$$

- Intersection

$$A \cap B = \{x: x \in A \text{ and } x \in B\}.$$

For example,

$$\{1, 3, 9\} \cap \{3, 5, 7\} = \{3\}$$

- Difference

$$A - B = \{x: x \in A \text{ or } x \notin B\}.$$

For example,

$$\{1, 3, 9\} - \{3, 5, 7\} = \{1, 9\}$$

- Disjoints

Two sets are **disjoint** if they have not element in common, that is, if their intersection is empty.

Set Operations

● Power set

A **power set** is a set of all subsets of a given set A and is denoted by 2^A .

Thus, $2^A = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

The number of elements = 2^n ,
i.e. $2^3 = 8$.

● Properties of set operations

Idempotency	$A \cup A = A, \quad A \cap A = A$
Commutativity	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Absorption	$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$
DeMorgan's Laws	$A - (B \cup C) = (A - B) \cap (A - C)$ $A - (B \cap C) = (A - B) \cup (A - C)$

1.2 Relation

Review

A relation is a set of ordered pairs. Given sets A and B , $R \subseteq A \times B$ is a binary relation from A to B . So, relations are also a set.

Example

Relation:

$$\{(1, 2), (2, 4), (3, 5), (2, 6), (1, -3)\}$$

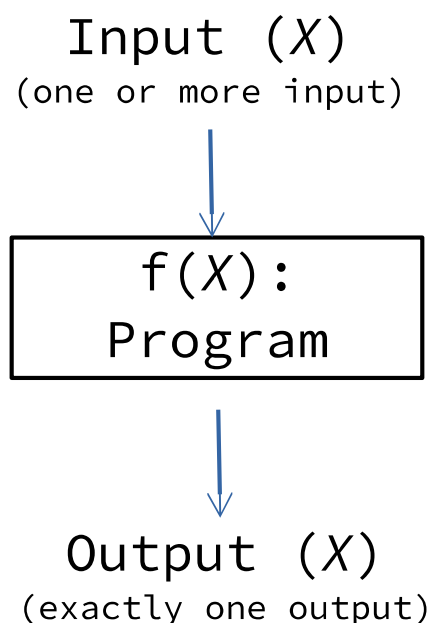
1.3 Functions

Review

A *Function* relates (maps) input to output.

Definition = $f(X \rightarrow Y)$
= X and Y are non-empty sets
= Can be *one to one* or *many to one* but *not one to many*.

In terms of *input/output*.



1.4 Alphabet and Language

Alphabet: a finite set of non-empty symbols.

Example:

- * $\Sigma = \{a, b, c, \dots, z\}$ the set of all lower case letters.
- * The set of all ASCII Characters.

String: finite sequence of symbols from an alphabet Σ ,
e.g. 0011001

Empty String: The string with zero occurrence of symbols from Σ .

Length of String: $|w|$ is the length of string w .
e.g. $|0110| = 4$, $|| = 0$

Powers of an Alphabet:

Σ^k is the set of strings of length k with symbols from Σ .

Example:

$$\begin{aligned}\Sigma &= \{0, 1\} \\ \Sigma^1 &= \{0, 1\} \\ \Sigma^2 &= \{00, 01, 10, 11\} \\ \Sigma^0 &= \{\}\end{aligned}$$

Question: How many strings are there in Σ^3 ?

The set of all strings over Σ is denoted Σ^* .

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$$

also,

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

Which indicates, $\Sigma^* = \Sigma^+ \cup \{\}$.

Concatenation:

If x and y are strings, then xy is the string obtained by placing a copy of y immediately after a copy of x .

Example:

$$x = a_1a_2a_3\dots a_m$$

$$y = b_1b_2b_3\dots b_n \text{ then,}$$

$$xy = a_1a_2a_3\dots a_m b_1b_2b_3\dots b_n$$

Languages:

If Σ is an alphabet, and $L \subseteq \Sigma^*$ then L is a language.

Example:

- The set legal English words
- The set of legal C programs.
- The set of strings consisting of n 0's followed by n 1's.

$$\{\emptyset, 01, 0011, 000111, \dots\}$$

The language $\{\}$ consisting of the empty string.